Evaluating Limits

ANSWER KEY

Section 1: Direct Substitution and Algebraic Techniques

1 Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3}$.

a. 3/2 ✓

b. 0

c. 1

d. Does not exist

Indeterminate $0/0$. Factor: $\frac{(x-3)(x+3)}{(x-3)(x+1)} = \frac{x+3}{x+1}$. Substitute $x = 3$: $\frac{3+3}{3+1} = \frac{6}{4} = \frac{3}{2}$.

2 Evaluate $\lim_{x \to 0} \frac{\sqrt{x+4} - 2}{x}$.

a. 1/4 ✓

b. 1/2

c. 0

d. 1/8

Multiply numerator and denominator by conjugate $(\sqrt{x+4} + 2)$: $\frac{(x+4)-4}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2}$. At $x = 0$, $\frac{1}{2+2} = \frac{1}{4}$.

3 Evaluate $\lim_{x \to 1} \frac{\frac{1}{x+1} - \frac{1}{2}}{x - 1}$.

a. $-1/4$ ✓

b. 1/4

c. $-1/2$

d. 0

Combine fractions: $\frac{2-(x+1)}{2(x+1)(x-1)} = \frac{1-x}{2(x+1)(x-1)} = \frac{-1}{2(x+1)}$. At $x = 1$, $\frac{-1}{2(2)} = -\frac{1}{4}$.

Section 2: Trigonometric Limits and the Squeeze Theorem

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4 Evaluate $\lim_{x \to 0} \frac{\sin(5x)}{3x}$.

a. 5/3 ✓

b. 3/5

c. 1

d. 0

Rewrite as $\frac{5}{3} \cdot \frac{\sin(5x)}{5x}$. Since $\lim_{u \to 0} \frac{\sin u}{u} = 1$, the limit is $5/3$.

5 Evaluate $\lim_{x \to 0} \frac{1 - \cos(x)}{x^2}$.

a. 1/2 ✓

b. 0

c. 1

d. Does not exist

Multiply by $(1 + \cos x)/(1 + \cos x)$: $\frac{\sin^2 x}{x^2(1 + \cos x)} = \left(\frac{\sin x}{x}\right)^2 \frac{1}{1 + \cos x} \to 1^2 \cdot \frac{1}{2} = \frac{1}{2}$.

6 If $1 - \frac{x^2}{6} \le g(x) \le 1$ for all x near 0, find $\lim_{x \to 0} g(x)$.

a. 1 ✓

b. 0

c. 5/6

d. Cannot be determined

Since $\lim_{x \to 0} (1 - x^2/6) = 1$ and $\lim_{x \to 0} 1 = 1$, by the Squeeze Theorem, $\lim_{x \to 0} g(x) = 1$.

Section 3: Infinite Limits and Limits at Infinity

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7 Evaluate $\lim_{x\to\infty} \frac{4x^3-2x+7}{5x^3+3x^2-1}$.

a. $4/5$ ✓

b. 0

c. ∞

d. $7/(-1)$

Divide numerator and denominator by x^3: $\frac{4-2/x^2+7/x^3}{5+3/x-1/x^3} \to \frac{4}{5}$ as $x\to\infty$.

8 Evaluate $\lim_{x\to-\infty} \frac{\sqrt{9x^2+4}}{2x-1}$.

a. $-3/2$ ✓

b. $3/2$

c. $9/2$

d. $-9/2$

For $x<0$, $\sqrt{x^2}=|x|=-x$. Thus $\sqrt{9x^2+4}=-x\sqrt{9+4/x^2}$. Ratio becomes $\frac{-x\sqrt{9}}{2x}=-\frac{3}{2}$.

9 Find $\lim_{x\to2^+} \frac{3x-1}{x-2}$ and $\lim_{x\to2^-} \frac{3x-1}{x-2}$.

a. $+\infty$ **and** $-\infty$ ✓

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 5 and 5

Numerator approaches $3(2)-1=5>0$. As $x\to2^+$, $x-2\to0^+$, so ratio $\to+\infty$. As $x\to2^-$, $x-2\to0^-$, so ratio $\to-\infty$.

Section 4: Piecewise and One-Sided Limits

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10 Let $f(x)$ be defined as $x^2 + 2$ for $x < 1$ and $4 - x$ for $x \geq 1$. Evaluate $\lim_{x \to 1} f(x)$.

a. 3 ✓

b. 1

c. 2

d. Does not exist

Left limit: $\lim_{x \to 1^-} (x^2 + 2) = 3$. Right limit: $\lim_{x \to 1^+} (4 - x) = 3$. Since both equal 3, the two-sided limit is 3.

11 Evaluate $\lim_{x \to 0^+} \frac{|x|}{x}$ and $\lim_{x \to 0^-} \frac{|x|}{x}$.

a. 1 **and** -1 ✓

b. 1 and 1

c. -1 and -1

d. 0 and 0

For $x > 0$, $|x| = x$, so $x/x = 1$. For $x < 0$, $|x| = -x$, so $-x/x = -1$.

12 Find the value of k such that $\lim_{x \to 2} g(x)$ exists, where $g(x)$ is $kx - 3$ for $x \leq 2$ and $x^2 + 1$ for $x > 2$.

a. 4 ✓

b. 2

c. 5

d. 3

Require $\lim_{x \to 2^-} g(x) = \lim_{x \to 2^+} g(x) \Rightarrow 2k - 3 = 2^2 + 1 = 5 \Rightarrow 2k = 8 \Rightarrow k = 4$.

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